

SubterraView Project Requirements Summary

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Overview

SubterraView is an augmented reality and AI-based system designed to improve underground infrastructure management and worker safety. The platform aggregates data from public utility records, city infrastructure databases, and crowdsourced field reports to generate a continuously updated map of underground utilities. An AI validation engine cross-references incoming data, assigns confidence ratings, and resolves inconsistencies to ensure accuracy and reliability. SubterraView is intended for construction workers, utility technicians, emergency responders, and both public and private organizations, providing real-time underground visualization and hazard alerts directly through a smartphone interface.

Scopes of Work and Product

The scope of the work focuses on the digital detection, validation, visualization, and monitoring of underground infrastructure data within municipal utility management and safety coordination. Currently, underground infrastructure management relies on 811 notifications, fragmented utility databases, manual field verification, and surface paint markings, making the process time-consuming and prone to inconsistencies. The scope of the product includes the development of a digital mapping and safety coordination platform that integrates AI-based record validation, augmented reality visualization of subsurface utilities, crowdsourced reporting, and real-time hazard alerts. This allows construction crews to visualize underground infrastructure through AR, utility technicians to submit updates for improved accuracy, and emergency responders to access critical underground layouts for safer and more efficient decision-making.

Mandated Constraints

The project is governed by several key mandated constraints to ensure its success and safety:

- **811 Compliance:** SubterraView must comply with the federally mandated 811 call-before-you-dig system and provide users with access to 811 services before any excavation.
- **Mobile Platform Requirement:** The system must operate on iOS and Android devices to ensure accessibility for field workers and responders.
- **External Sensor Integration:** The system must integrate with external hardware sensors to detect hazards such as gas leaks and environmental risks.
- **Data Source Dependency:** The system must utilize and integrate data from external sources including city infrastructure databases, GIS systems, and crowdsourced inputs.
- **Mobile Feature Compatibility:** The system must be compatible with built-in mobile functionalities such as GPS, camera, messaging, and file storage.

Functional Requirements

- User Authentication: The system must allow users to create accounts and log in with a defined role that determines their access level within SubterraView.
- AR Visualization: The system must provide an augmented reality view that displays underground infrastructure for a selected location.
- Location Detection: The system must determine user location through GPS or manual input to retrieve accurate infrastructure data.
- Hazard Alerts: The system must notify relevant users of detected hazards in real time using connected sensors.
- Field Reporting: Users must be able to submit field reports to update or correct infrastructure data.
- 811 Integration: The system must provide users with direct access to 811 services for compliance with excavation regulations.
- Emergency Mode: The system must provide a dedicated emergency mode that displays a prioritized underground map for responders.
- Latency: AR visualization and hazard alerts must be delivered within a few seconds to ensure timely decision-making.
- Accuracy: The AI validation engine must maintain a high level of accuracy ($\geq 97\%$) when processing infrastructure data.
- Availability: The system must maintain high availability with minimal downtime to support continuous use.
- Robustness: The system must continue to function with limited capabilities during network disruptions by using cached data.
- Access Control: A role-based access control system must restrict access to data and features based on user roles.
- Data Privacy: Sensitive user and location data must be protected from unauthorized access.
- Data Integrity: Infrastructure and hazard data must be protected from unauthorized modification.
- Cross-Platform Support: The system must function across iOS and Android devices.
- Scalability: The system must support growth in users, data volume, and geographic regions.
- Ease of Use: The interface must be intuitive and allow users to complete tasks with minimal effort.
- Compliance: The system must follow regulations such as 811 requirements and applicable safety standards.
- Standards: The system must follow industry standards for data formatting and infrastructure mapping.

Each requirement is paired with a corresponding acceptance test that verifies whether the system meets its expected behavior.